Acid-Base: Acidosis vs Alkalosis Overview & Compensation

1. Henderson rule: pH ~ HCO3/CO2

WHAT YOU SEE

Top equation '↓PH, ↓HCO3 ↓CO2'

WHAT IT ENCODES

pH is governed by the ratio of HCO3- (metabolic, kidney) to PCO2 (respiratory, lung): pH □ HCO3/PCO2. Acidosis lowers pH; alkalosis raises it. This single ratio frames every acid-base problem.

BOARD TRAP

Look at the RATIO, not absolute values — a low HCO3 with proportionally low CO2 can keep pH near normal (compensation).

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2. Acidosis (left side)

WHAT YOU SEE

'ACIDOSIS' banner, dim/shaded left half

WHAT IT ENCODES

Acidosis = primary process lowering pH below 7.35. Metabolic acidosis = low HCO3; respiratory acidosis = high PCO2 from hypoventilation. The left 'dark' side groups acid-gaining states.

BOARD TRAP

An acidemic pH can result from either a metabolic OR respiratory primary — the PCO2 and HCO3 direction tells you which.

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3. Alkalosis (right side)

WHAT YOU SEE

'Alkalosis' banner, bright/sunny right half

WHAT IT ENCODES

Alkalosis = primary process raising pH above 7.45. Metabolic alkalosis = high HCO3 (vomiting, diuretics, hyperaldosteronism); respiratory alkalosis = low PCO2 from hyperventilation.

BOARD TRAP

Vomiting causes metabolic ALKALOSIS (loss of gastric H+/Cl-), a frequent exam point — not acidosis despite GI losses.

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4. Respiratory CO2 (RES label)

WHAT YOU SEE

'RES' / CO2 markers left, '↑CO2'

WHAT IT ENCODES

Respiratory component is set by PCO2: hypoventilation raises CO2 (respiratory acidosis), hyperventilation lowers it (respiratory alkalosis). Lungs adjust within minutes.

BOARD TRAP

PCO2 changes are the FAST respiratory limb; renal HCO3 compensation is slow (hours-days), so timing distinguishes acute from chronic.

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5. Metabolic HCO3 (couch)

WHAT YOU SEE

Orange coiled 'metabolic' couch, ↑/↓ HCO3

WHAT IT ENCODES

Metabolic component is bicarbonate, regulated by the kidney. Low HCO3 = metabolic acidosis; high HCO3 = metabolic alkalosis. Renal compensation takes 1-3 days.

BOARD TRAP

A pure metabolic disorder shifts HCO3 first; if PCO2 hasn't compensated 'appropriately,' a second respiratory disorder coexists.

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6. Hyperventilation (panting)

WHAT YOU SEE

Character panting/blowing, 'Hypervent'

WHAT IT ENCODES

Hyperventilation blows off CO2 → respiratory alkalosis (anxiety, salicylates, sepsis, high altitude, PE). Also serves as compensation FOR metabolic acidosis (Kussmaul breathing).

BOARD TRAP

Salicylate toxicity causes BOTH primary respiratory alkalosis AND metabolic (anion-gap) acidosis — a classic mixed disorder.

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7. Tickets cone / pH range

WHAT YOU SEE

Tickets cone, normal pH '7.35–7.45'

WHAT IT ENCODES

Normal arterial pH is 7.35-7.45 (midpoint 7.40). Below 7.35 = acidemia, above 7.45 = alkalemia. Normal HCO3 22-28 mEq/L; normal PCO2 35-45 mmHg.

BOARD TRAP

pH 7.40 with abnormal HCO3 AND PCO2 means a fully compensated or mixed disorder — never call it 'normal' without checking gases.

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8. HCO3 22-28 normal

WHAT YOU SEE

'22-28' and '22-2' on ticket dispenser

WHAT IT ENCODES

Normal serum bicarbonate is 22-28 mEq/L. Drops in metabolic acidosis, rises in metabolic alkalosis or as chronic respiratory acidosis compensation.

BOARD TRAP

A high HCO3 may be primary metabolic alkalosis OR renal compensation for chronic respiratory acidosis — check pH and PCO2 to tell apart.

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9. PCO_2 35-45 normal

WHAT YOU SEE

'7.35' and '7.45' / pH range markers at base

WHAT IT ENCODES

Normal PCO_2 is 35-45 mmHg. The respiratory system can change it in minutes. Winter's formula predicts expected PCO_2 in metabolic acidosis: $1.5 \times HCO_3 + 8 \pm 2$.

BOARD TRAP

If measured PCO_2 differs from Winter's-predicted value, a concurrent respiratory disorder exists — a high-yield mixed-disorder clue.

10. 3-Ks compensation cue

WHAT YOU SEE

'3S = KS' annotation near diver

WHAT IT ENCODES

Compensation moves in the SAME direction as the primary disorder: in metabolic acidosis (low HCO_3), the lung lowers PCO_2; in metabolic alkalosis, PCO_2 rises. Compensation never fully corrects pH or overcorrects.

BOARD TRAP

Overcorrection (pH crossing to the opposite abnormal side) means a SECOND primary disorder, not just compensation.

11. Same direction principle

WHAT YOU SEE

Bottom banner 'Same Direction Simple ACID Base'

WHAT IT ENCODES

Key rule: in a simple disorder, HCO_3 and PCO_2 move in the SAME direction. Metabolic acidosis: both fall. Respiratory acidosis: both rise. Opposite movements signal a mixed disorder.

BOARD TRAP

If HCO_3 and PCO_2 move in OPPOSITE directions, it is always a mixed acid-base disorder — a guaranteed board discriminator.

12. Metabolic alkalosis (couch R)

WHAT YOU SEE

Orange couch right, 'metabolic $\uparrow HCO_3$'

WHAT IT ENCODES

Metabolic alkalosis: elevated HCO_3 from vomiting/NG loss, diuretics, mineralocorticoid excess. Saline-responsive (urine Cl <20, volume depletion) vs saline-resistant (urine Cl >20, hyperaldosteronism).

BOARD TRAP

Urine chloride distinguishes the two: low = vomiting/diuretics (give saline); high = hyperaldosteronism (saline won't help).

13. Respiratory alkalosis (hyperv R)

WHAT YOU SEE

Right side fan/blower, '$\downarrow CO_2$ Hyperv'

WHAT IT ENCODES

Respiratory alkalosis from hyperventilation: anxiety, pain, sepsis, salicylates, pregnancy, high altitude, PE. PCO_2 falls; pH rises. Renal compensation lowers HCO_3 over days.

BOARD TRAP

Early sepsis/PE often presents with primary respiratory alkalosis — don't dismiss tachypnea as anxiety in an ill patient.

14. Water pHun Park theme

WHAT YOU SEE

'Water pHun PARK' sign

WHAT IT ENCODES

The water-park scene anchors the pH axis: dark/cold left = acidosis, bright/warm right = alkalosis, with respiratory (CO_2) and metabolic (HCO_3) characters on each side. A mnemonic map for stepwise interpretation.

BOARD TRAP

Always interpret in order: pH → primary disorder → compensation adequacy → anion gap → delta-delta — skipping steps misses mixed disorders.